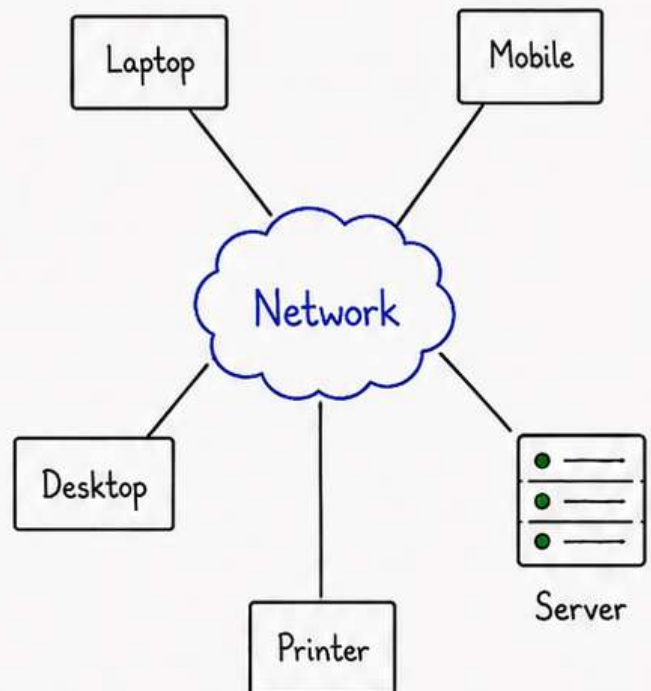


COMPUTER NETWORKS

— Page 1 —

1. What is Computer Network ?

A computer network is a collection of interconnected devices that can communicate and share data, resources and services.



2. Why do we need Computer Networks ?

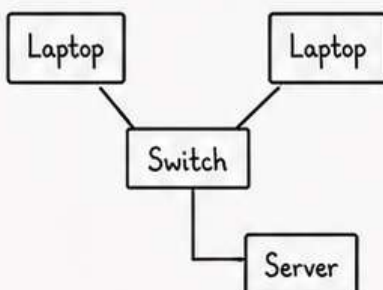
- ① Share files, data and resources
- ② Communication (Email, Chat, Video Call)
- ③ Internet access
- ④ Centralized management and security
- ⑤ Cost and time saving

3. Types of Networks

LAN

(Local Area Network)

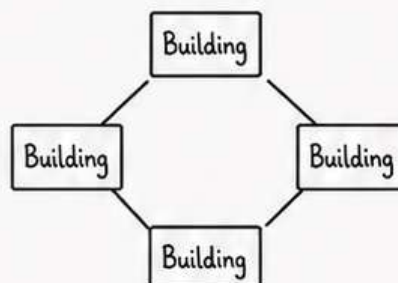
- Covers small area like home, office, building.
- High speed
- Privately owned
- Example : Office LAN, Home Wi-Fi



MAN

(Metropolitan Area Network)

- Covers a city or metropolitan area.
- Higher than LAN but lower than WAN.
- Example : Cable TV network in a city



WAN

(Wide Area Network)

- Covers large geographical area like country or world.
- Lower speed than LAN but covers long distance.
- Example : Internet



OSI MODEL

(Open Systems Interconnection Model)

— Page 2 —

→ OSI Model is a conceptual framework that **standardizes** how data is **communicated** over a network.

Why OSI Model ?

- Helps in understanding network flow
- Troubleshooting becomes easy
- Vendor independent model

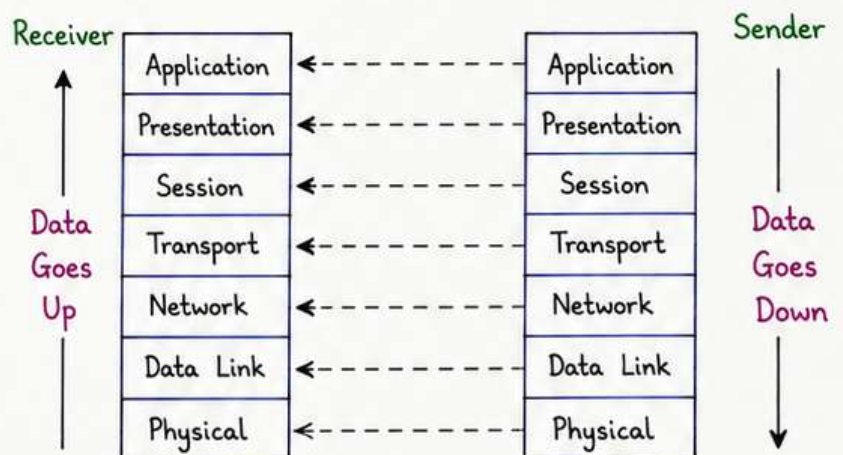
Layer	Layer Name	Main Function
⑦	Application Layer	Provides network services to user applications. (HTTP, FTP, SMTP, DNS)
⑥	Presentation Layer	Handles data format, encryption, decryption, compression.
⑤	Session Layer	Establishes, manages and terminates sessions between applications.
④	Transport Layer	Ensures end-to-end communication, reliability, segmentation. (TCP, UDP)
③	Network Layer	Logical addressing and routing of packets. (IP)
②	Data Link Layer	Framing, MAC addressing, error detection. (Switches)
①	Physical Layer	Transmits raw bits over physical medium. (Cables, Hubs)

Mnemonic to remember :

A Application
 P Presentation
 S Session
 T Transport
 N Network
 D Data Link
 P Physical

APSTNDP

How data travels (Encapsulation) ?



Each layer provides **services** to the layer above it and uses the **services** of the layer below it.

TCP/IP MODEL

— Page 3 —

→ TCP/IP Model is the practical implementation model used in the Internet. It has 4 layers.

Why TCP/IP Model ?

- Simpler and easier to understand
- Used in real-world Internet communication
- Protocol independent
- Supports internetworking

Layer	Layer Name	Main Function	Examples / Protocols
④	Application Layer	Provides network services to applications. (User level)	HTTP, HTTPS, FTP, SMTP, DNS, DHCP, Telnet
③	Transport Layer	End-to-end communication, reliability, error control, flow control.	TCP, UDP
②	Internet Layer	Logical addressing and routing of packets across networks.	IP (IPv4, IPv6), ICMP, ARP
①	Network Access Layer	Transmits data over the physical network.	Ethernet, Wi-Fi, PPP

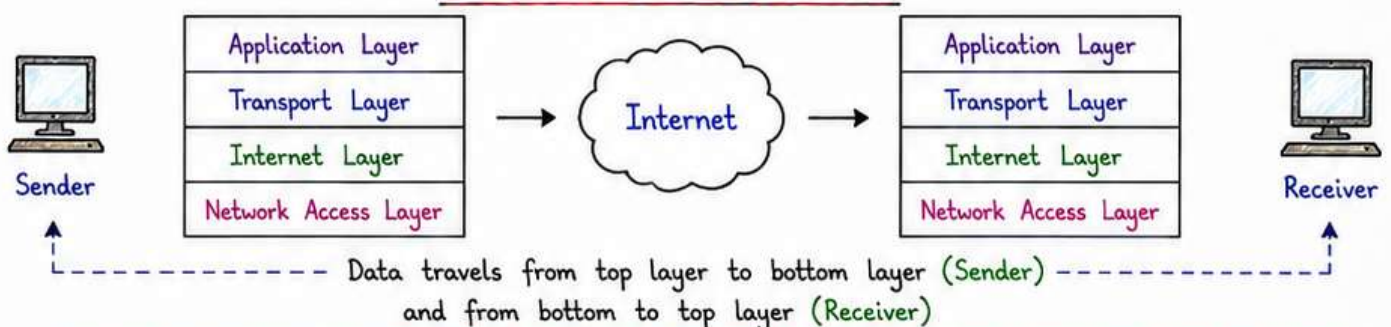
OSI vs TCP/IP Model (Mapping)

OSI Model (7 Layers)	TCP/IP Model (4 Layers)
7. Application 6. Presentation 5. Session	Application Layer
4. Transport	Transport Layer
3. Network	Internet Layer
2. Data Link 1. Physical	Network Access Layer

Key Points

- ① TCP/IP model is based on protocol suite (TCP/IP).
- ② OSI model is a reference model, TCP/IP is a protocol model.
- ③ OSI has 7 layers, TCP/IP has 4 layers.
- ④ TCP/IP is simpler and widely used in real world.

Packet Flow in TCP/IP Model



In Short : TCP/IP model helps devices communicate over the Internet by breaking data into packets, sending them, and reassembling at receiver.

TCP vs UDP

— Page 4 —

1. What are TCP and UDP ?

- **TCP** (Transmission Control Protocol) is a **connection-oriented**, **reliable** protocol.
- **UDP** (User Datagram Protocol) is a **connectionless**, **faster** protocol.

TCP

- ✓ Reliable
- ✓ Slower
- ✓ Connection-Oriented

UDP

- ✓ Faster
- ✓ Less Reliable
- ✓ Connectionless

2. TCP vs UDP (Comparison)

Feature	TCP	UDP
Connection	Connection - Oriented	Connectionless
Reliability	Reliable (ACK, Retransmission)	Less Reliable (No ACK)
Speed	Slower	Faster
Data Order	Maintains Order	No Order Guarantee
Error Checking	Yes	Yes (Basic)
Flow Control	Yes	No
Use Cases	Web, Email, File Transfer, Online Banking	Video Streaming, Gaming, VoIP, Live Broadcast

3. Real World Examples

TCP Used In

- Web Browsing (HTTP/HTTPS)
- Email (SMTP, IMAP, POP3)
- File Transfer (FTP)
- Online Banking
- Remote Login (SSH)

UDP Used In

- Video Streaming (YouTube Live)
- Online Gaming
- Voice / Video Calls
- Live Broadcast
- DNS (Sometimes)

Choose **TCP** when reliability matters.
Choose **UDP** when speed matters. 😊

IP Address & MAC Address

— Page 5 —

1. IP Address

- IP (Internet Protocol) Address is a **unique address** assigned to a device on a network.
- It helps in identifying devices **across different networks**.

Types of IP Address

- **IPv4** : 32-bit address
Example : 192.168.1.1
- **IPv6** : 128-bit address
Example : 2001:0db8:85a3::8a2e:0370:7334

2. MAC Address

- MAC (Media Access Control) Address is a **unique hardware address** assigned to a network interface (NIC).
- It is used to identify devices **within the local network (LAN)**.

MAC Address Format (48-bit)

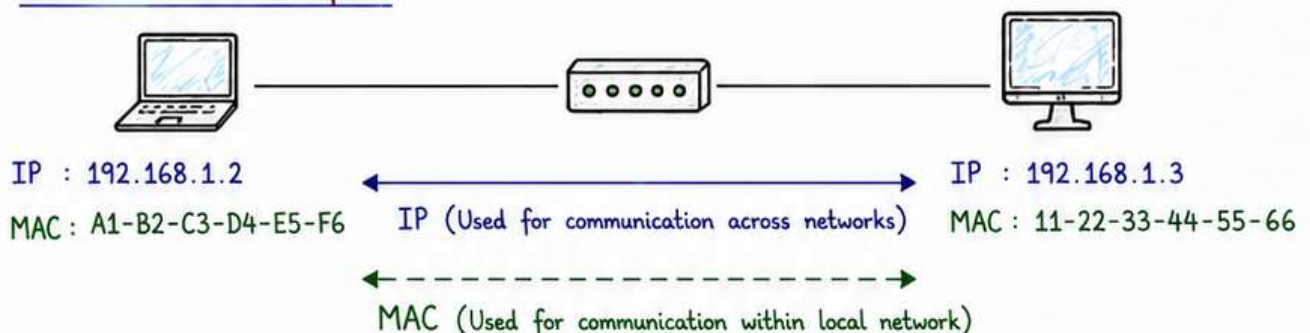
A1 - B2 - C3 - D4 - E5 - F6

Example : A1-B2-C3-D4-E5-F6

3. IP vs MAC

Feature	IP Address	MAC Address
Full Form	Internet Protocol Address	Media Access Control Address
Layer	Network Layer (Layer 3)	Data Link Layer (Layer 2)
Length	32-bit (IPv4) / 128-bit (IPv6)	48-bit
Address Type	Logical Address	Physical Address
Changeable	Yes (Can be changed)	No (Hardcoded in NIC)
Used For	Communication between networks	Communication within local network
Example	192.168.1.1	A1-B2-C3-D4-E5-F6

4. How it Works (Simple)



DNS (Domain Name System)

— Page 6 —

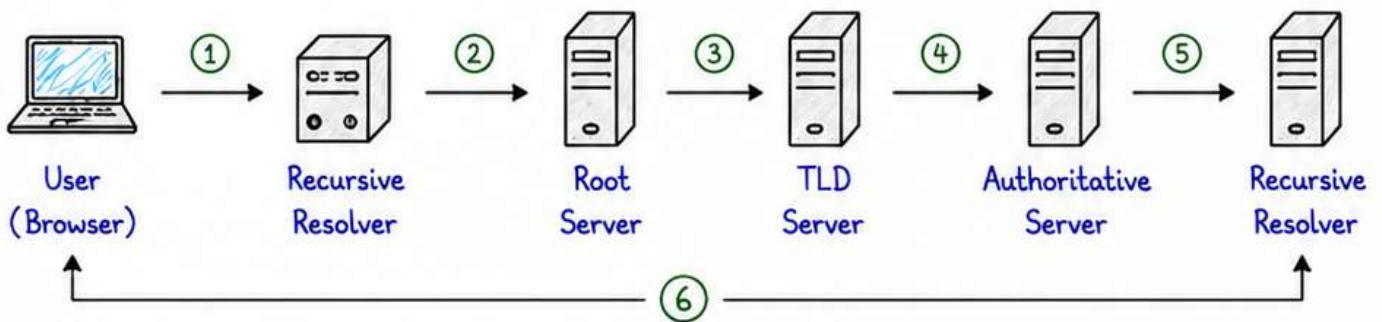
1. What is DNS ?

- DNS is like the "phonebook of the Internet".
- It converts domain names (like `google.com`) into IP addresses (like `142.250.190.14`).

Why DNS is Needed ?

- ✓ Easy for humans to remember names.
- ✓ IP addresses are hard to remember.
- ✓ If IP changes, only DNS record needs to update.

2. DNS Resolution Process



- ① User enters `www.google.com` in browser.
- ② Recursive Resolver checks its `cache`. If not found, it asks Root Server.
- ③ Root Server tells the address of TLD Server (`.com`).
- ④ TLD Server gives the address of Authoritative Server for `google.com`.
- ⑤ Authoritative Server returns the IP address of `google.com`.
- ⑥ Recursive Resolver sends the IP back to the browser.

This process is very fast (in ms) due to **caching**.

3. Example

You type `www.google.com` → Gets converted to `142.250.190.14`
Then browser connects to this IP and loads the website.

DNS makes Internet easy for humans ! 😊

HTTP vs HTTPS

— Page 7 —

1. What are HTTP and HTTPS ?

- **HTTP** (HyperText Transfer Protocol) is used to **transfer data** between client and server.
- **HTTPS** (HTTP Secure) is the **secure** version of HTTP. It uses **encryption** (SSL/TLS).

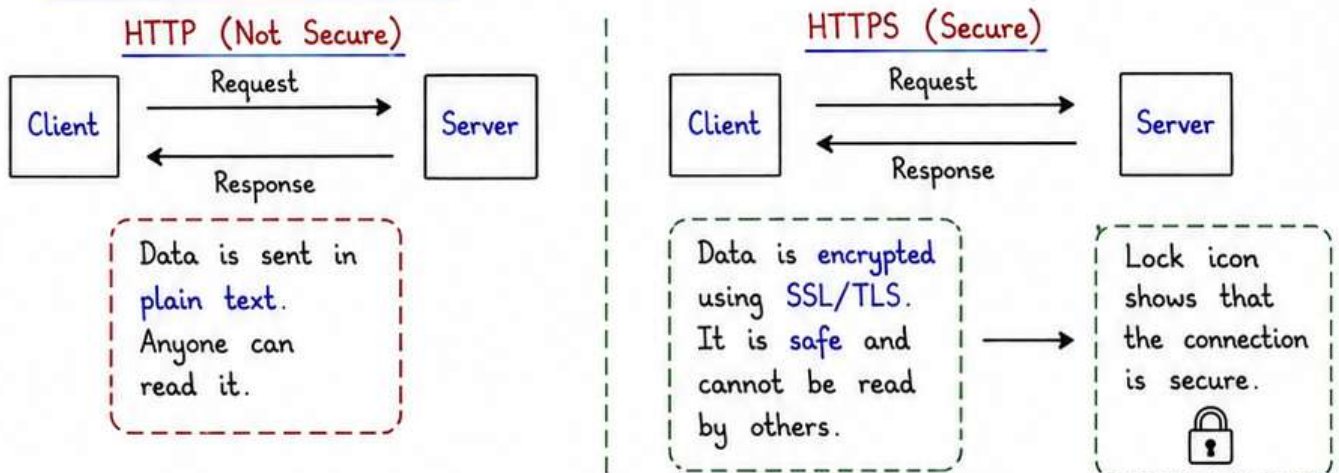
Why HTTPS ?

- Encrypts data
- Protects from hackers
- Builds trust (lock in browser)

2. HTTP vs HTTPS (Comparison)

Feature	HTTP	HTTPS
Full Form	HyperText Transfer Protocol	HyperText Transfer Protocol Secure
Security	Not Secure (Data in Plain Text)	Secure (Data is Encrypted)
Port	80	443
Protocol	Uses TCP	Uses TCP + SSL/TLS
Encryption	No	Yes
URL Format	http://example.com	https://example.com
Lock Icon	No Lock	Lock Present
Use	Less Secure Websites	Secure Websites (Banking, Payment, Email)

3. HTTP vs HTTPS (Flow)



In Short : Always prefer **HTTPS** over HTTP for secure communication.

How a Browser Opens a Website ?

— Page 8 —

① You Type the URL

- You enter a website URL in the browser.
Example: `google.com`

② DNS Resolution

- Browser asks DNS server for IP address of the domain.
- It returns the IP (e.g., `142.250.190.78`).

③ TCP Connection

- Browser creates a TCP connection with the server (3-way handshake).

④ TLS / SSL Handshake (For HTTPS)

- Browser and server verify each other and create a secure connection.

⑤ HTTP Request

- Browser sends an HTTP request to the server.

⑥ Server Processing

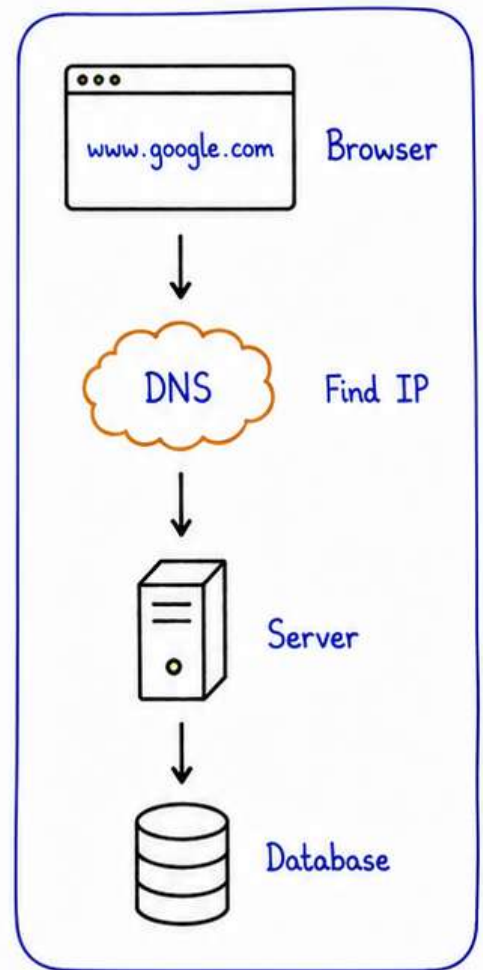
- Server processes the request.
It may fetch data from database.

⑦ Response to Browser

- Server sends back the response (HTML, CSS, JS, Images, etc.).

⑧ Browser Renders the Page

- Browser reads the response and displays the website on your screen.



Why this flow?

- ✓ DNS helps find the server.
- ✓ TCP ensures reliable connection.
- ✓ HTTPS keeps data secure.
- ✓ Server processes your request.
- ✓ Response is shown in browser.

Example:

When you open `google.com`, this is the complete journey happening in milliseconds.

It feels instant for us, but behind the scenes, many steps happen! 😊

LOAD BALANCER

— Page 9 —

1. What is Load Balancer ?

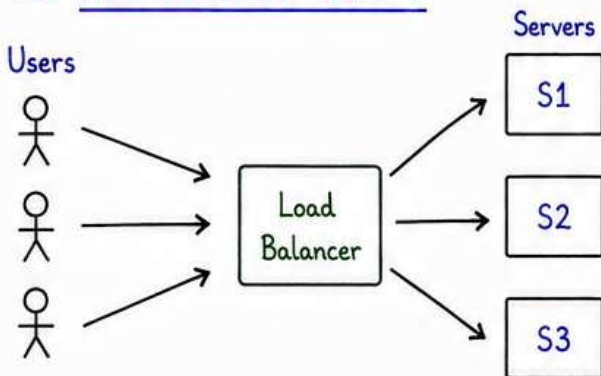
A Load Balancer distributes incoming **network traffic** across multiple servers to ensure no single server is **overloaded**.

It improves **performance**, **reliability** and **availability**.

2. Why is it Needed ?

- Prevents server from getting overloaded.
- Improves response time.
- Ensures high availability.
- Helps in scaling applications.

3. How Does It Work ?



- ① Users send request to the application.
- ② Request first goes to **Load Balancer**.
- ③ Load Balancer decides which server can handle the request.
- ④ It forwards the request to that server.
- ⑤ Server processes the request and sends response back to the user (through **Load Balancer**).

4. Load Balancing Algorithms (Common)

Algorithm	How It Works
Round Robin	Requests are distributed one by one in a sequential order to each server.
Least Connections	Request is sent to the server with the least number of active connections.
IP Hash	Uses client IP address to always send requests to the same server.
Least Response Time	Sends request to the server with the least response time.

5. Benefits

- ✓ Better performance
- ✓ High availability
- ✓ Fault tolerance
- ✓ Scalability
- ✓ Efficient resource utilization

Load Balancer acts like a **smart traffic police** that controls and manages the incoming requests.



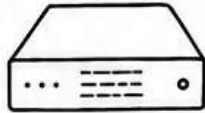
LOAD BALANCER

— Page 10 —

1. Types of Load Balancers

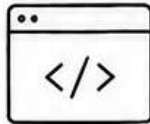
• Hardware Load Balancer

A physical device that distributes traffic.



• Software Load Balancer

A software program that runs on a server.



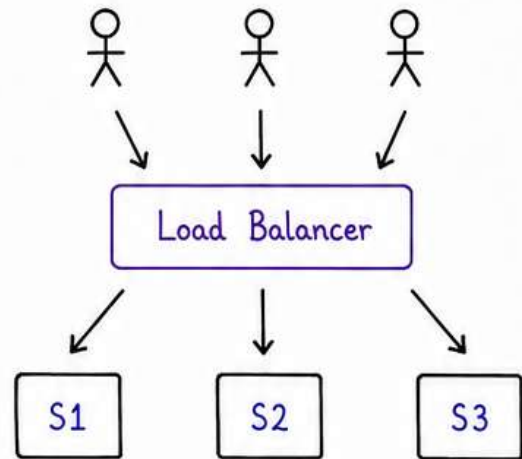
• Cloud Load Balancer

A managed service provided by cloud platforms.



2. Where is it Placed ?

It is placed between users and servers.

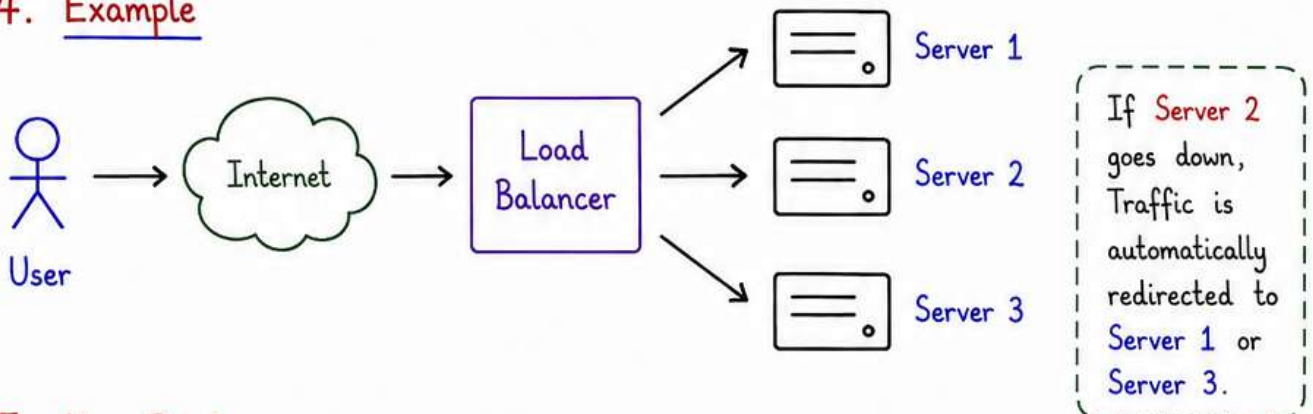


3. Health Check



- Load Balancer continuously checks the health of each server.
- If a server is down or not responding, it stops sending traffic to it.
- Traffic is sent only to healthy servers.
- This ensures smooth service without interruption.

4. Example



5. Key Points

✓
Distributes traffic evenly across servers.

✓
Improves performance and reliability.

✓
Prevents any single server from getting overloaded.

✓
Ensures high availability of applications.

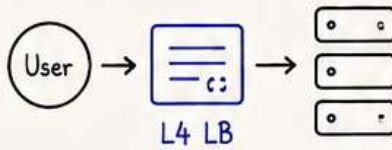
LOAD BALANCER

— Page 11 —

1. Types (Deployment)

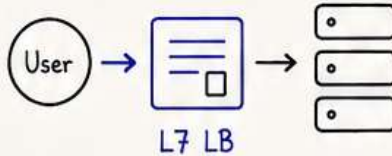
→ Layer 4 (Transport Layer)

Works with IP and port.
Faster, less inspection.



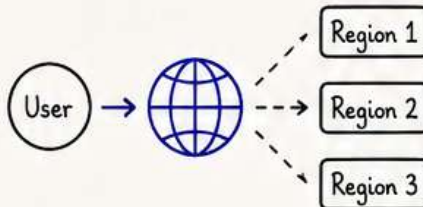
→ Layer 7 (Application Layer)

Works with HTTP/HTTPS,
inspects content.
Smarter routing.



→ Global Load Balancer

Distributes traffic across
regions (for global apps).



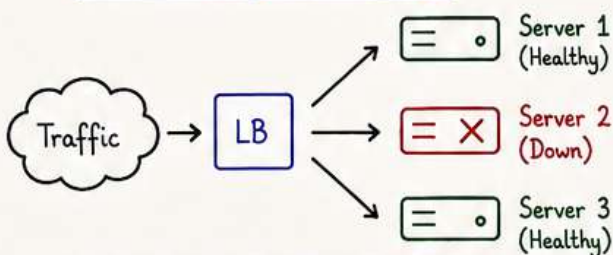
2. Functions

- ✓ Distributes incoming traffic.
- ✓ Monitors server health.
- ✓ Routes traffic to healthy servers.
- ✓ Handles failover.
- ✓ Improves performance and reliability.

3. Common Load Balancing Algorithms

Algorithm	How It Works	Best For
• Round Robin	Requests are distributed one by one in a sequential order.	All servers have same capacity.
• Least Connections	Request is sent to the server with the least number of active connections.	Servers have varying loads.
• IP Hash	Uses client IP to always send requests to the same server.	Session persistence required.
• Least Response Time	Sends traffic to the server with the lowest response time.	Performance critical applications.

4. Failover (High Availability)



If Server 2 is down, traffic is automatically sent to other healthy servers. ✓

5. Key Takeaway



Load Balancer is like a traffic cop for your application.

- ★ It keeps everything balanced.
- ★ It keeps servers healthy.
- ★ It keeps users happy. 😊

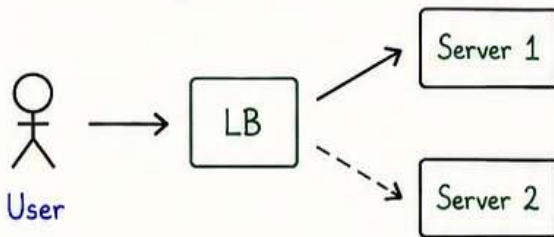
★ Balanced load = Happy servers + Happy users! 😊

LOAD BALANCER

— Page 12 —

1. Session Persistence

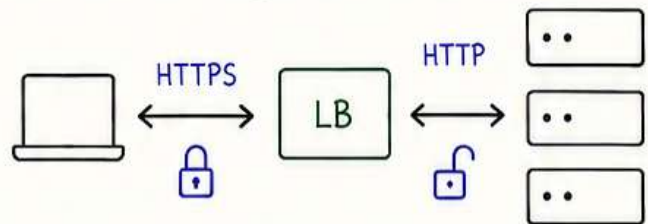
It makes sure a user's requests go to the **same server** during a session.



All requests in a session go to **Server 1**.

2. SSL Termination

Load Balancer handles **SSL** encryption and forwards requests to servers without encryption.



Offloads encryption work from servers (better performance).

3. Real-World Uses

Web Apps



Distributes user traffic across web servers.

E-commerce



Handles high traffic during sales and offers.

APIs



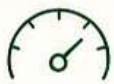
Balances requests across multiple API servers.

Streaming



Distributes viewer requests for smooth streaming.

4. Benefits



Better Performance



High Availability

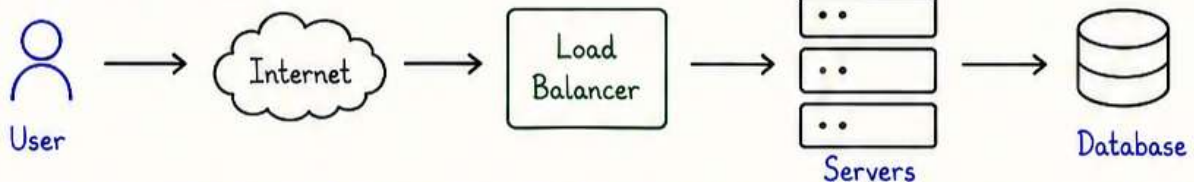


Easy Scalability



Efficient Resource Usage

5. Simple Flow Recap

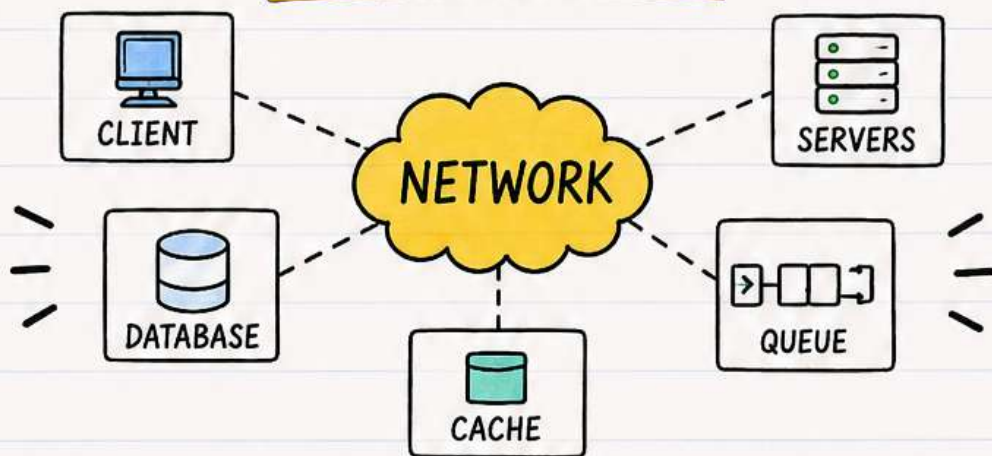


User request comes → LB decides → Server handles → Response goes back! 😊

♡ ≡ FOLLOW ≡ ♡ → FOR ← PART 2 ≡

MORE ADVANCED TOPICS,
EXAMPLES & REAL-WORLD PROJECTS AHEAD!

≡ COMMENT ≡ ☆
CN NOTES
↓
FOR PDF NOTES



☆ ≡ FOR PART 2 ≡ ☆
TARGET IS 10K ♡